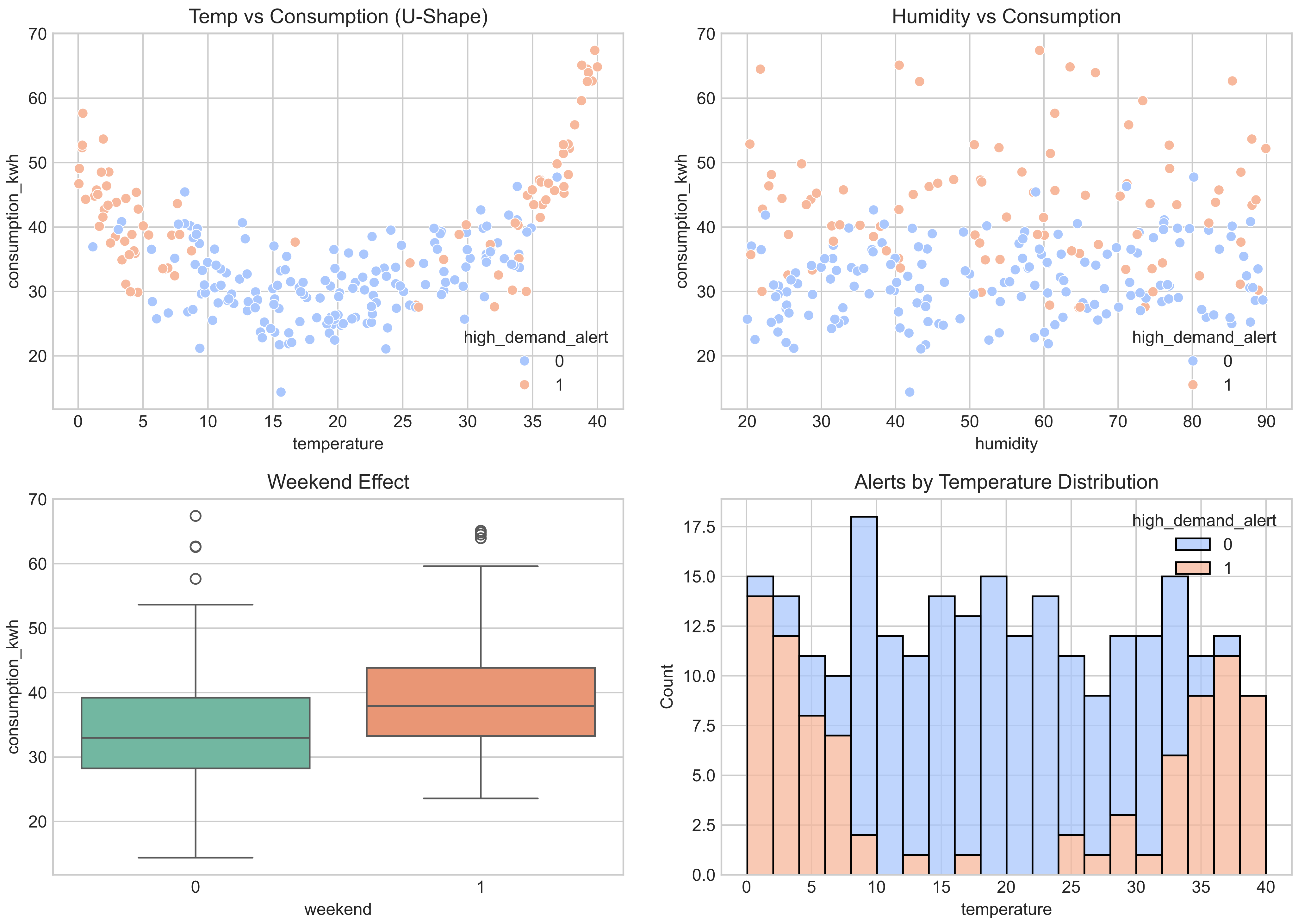
Data Exploration

SAMPLE SIZE & STRUCTURE

• Train Set: 250 independent days
• Test Set: 120 independent days
• Features: Temperature, Humidity, Weekend
• Target 1 (Reg): Consumption (kWh)
• Target 2 (Class): High Demand Alert (0/1)

SUMMARY STATISTICS

• Temperature Range: 0.06°C to 39.99°C
• Humidity Range: ~20% to ~90%
• Highly non-linear 'U-Shape' behavior
  observed in Temperature vs Consumption.
• Demand Baseline Shift observed on weekends.



Methodology and Results

METHODOLOGY:
• Modeled non-linear temperature effects using Cubic B-Splines (Cox-de Boor algorithm, m=3 internal knots).
• Task 1 mapped via custom Ordinary Least Squares (OLS). Task 2 mapped via Logistic Regression.

ASSUMPTIONS:
• Boundary knots clamped strictly at 0°C and 40°C to maintain stability at test-set extremes.
• B-splines natively form a Partition of Unity (sum to 1), requiring the omission of a global intercept.
• Humidity and Weekend status assumed as simple additive, linear effects.

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EXPLICIT MODELS:
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Task 1 (Regression):
Y = 47.00*B_1(T) + 32.78*B_2(T) + 29.65*B_3(T)
    + 21.47*B_4(T) + 34.49*B_5(T) + 35.35*B_6(T) + 64.65*B_7(T)
    + 0.04*H + 3.22*W


Task 2 (Classification):
P(C=1) = 1 / (1 + exp(-Z))
Z = 1.85*B_1(T) + 3.18*B_2(T) + -4.62*B_3(T)
    + -4.72*B_4(T) + -1.18*B_5(T) + -1.25*B_6(T) + 11.12*B_7(T)
    + 0.01*H + 0.27*W
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TASK 1 PERFORMANCE:
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MSE: 17.612
R²:  0.791
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TASK 2 PERFORMANCE:
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Log Loss: 0.314
AUC:      0.928
Accuracy: 0.868
F1 Score: 0.795
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Visualization: Test Set Predictions & Model Behavior